

SMART INDIA HACKATHON 2026



TeamSN

- **Problem Statement ID** – SIH26167
- **Problem Statement Title**- SatQuery AI - An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries
- **Theme**- Space Technology
- **PS Category**- Software

Team ID-

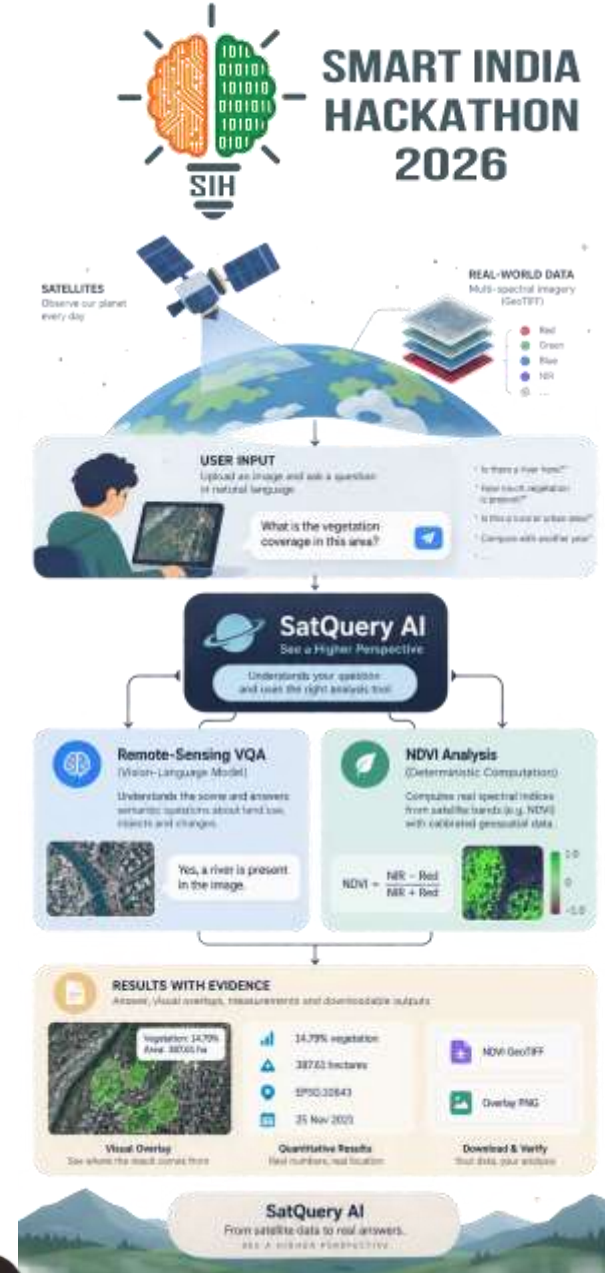
- **Team Name (Registered on portal) : TeamSN**



SATQuery AI

❖ Proposed Solution

- **Ask satellite images questions in plain English.** Upload a scene, ask what you want to know - no remote-sensing expertise required.
- **It understands what it sees.** A remote-sensing-trained vision-language model answers questions about land use, objects, rural vs. urban, and scene comparisons.
- **Specialized, not retrained from scratch.** A lightweight LoRA adaptation tunes the model for our exact use case - cheap to train, sharper on our workflow.
- **Real math for real numbers.** Quantitative questions get computed, not guessed. Our NDVI pipeline reads actual Red/NIR bands pixel by pixel using the standard vegetation-index formula.
- **Every answer comes with receipts.** Results include the exact pixels flagged, a visual overlay, calculated area, and a downloadable file - so you can see exactly where a number came from.
- **Coordinates, not just pixels.** Every scene keeps its real geographic footprint, so "this area" means an actual place on Earth.
- **Fully auditable.** Each run logs inputs, thresholds, calibration, and pixel-level data for independent verification.
- **One tool per job.** VQA and NDVI run as distinct modes now; the goal is a router that sends each question to the right specialist instead of one model doing everything.
- **Built to expand.** Same architecture, room to add grounding, change detection, and optical-SAR fusion later - without rebuilding what works.



TECHNICAL APPROACH

AI / Vision-Language Model: AdaptLLM Remote-Sensing Qwen2-VL-2B-Instruct, based on the Qwen2-VL architecture, runs locally for visual question answering on remote-sensing imagery.

Model Adaptation: LoRA / PEFT is used for parameter-efficient domain adaptation. The foundation model remains largely frozen while lightweight adapter parameters are trained for SatQuery's target remote-sensing VQA behaviour.

ML Framework: PyTorch + Hugging Face Transformers power local model loading, inference and adapter training. The working baseline runs in **BF16 without quantization**.

Backend: Python + FastAPI connect the browser interface to a serial local worker responsible for VLM inference, geospatial ingestion, deterministic analysis and result generation.

Geospatial Engine: Multispectral GeoTIFF imagery is retained with its geospatial metadata and calibrated bands. Raster processing supports CRS, geographic bounds, pixel validity, quality masking, overlays and spatial-area calculations.

Spectral Computation: NDVI is calculated directly from calibrated **NIR and Red bands** using $(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$. Thresholding, valid-pixel filtering and area calculations run deterministically on the **CPU**, independent of VLM inference.

Evidence & Outputs: SatQuery generates inspectable **NDVI overlays**, **pixel statistics**, **projected area measurements**, **geospatial metadata**, **downloadable GeoTIFF outputs** and **structured JSON run records** for reproducibility and verification.

Local Deployment: The prototype operates **locally on the user's laptop through Ubuntu/WSL**, keeping model inference and analysis on-device rather than depending on a cloud inference API.

Qwen2-VL-2B Local Baseline
 6.19 s model load • 4.3 GiB peak PyTorch GPU allocation
 2.93 s descriptive query • 0.36 s short VQA query
 BF16 • No quantization • Local GPU

Measured engineering runs; latency values are not benchmark-wide averages.



FEASIBILITY AND VIABILITY



Feasibility

Core stack combines a remote-sensing VLM, LoRA adaptation, and deterministic geospatial math - VQA and calibrated NDVI have been validated in early experiments

Runs on an 8GB consumer GPU (~4.3 GiB peak, BF16) - no enterprise hardware needed

AI reasoning and numerical computation stay separate, so new tools can be added without rebuilding the system

Viability

Works on standard multispectral GeoTIFF, fitting existing satellite workflows

Open-source stack (Python, FastAPI, PyTorch) - no proprietary lock-in

Designed to scale in stages: VQA → spectral indices → grounding → temporal → optical/SAR

Challenges

Remote-sensing data varies across sensors, seasons, and regions - real generalization needs rigorous benchmarking, not single-scene results

VLMs can't be trusted for precise numbers - quantitative tasks route to deterministic tools instead

Temporal and SAR fusion require registration and sensor-aware preprocessing beyond current scope

Use Cases

Agriculture (vegetation/spectral analysis) · Urban & environmental monitoring (land cover, development) · Disaster & change analysis (future temporal/SAR specialists)

Addressing the Challenges

Deterministic math for anything computable; VLM handles semantic reasoning only

Full traceability - thresholds, calibration, pixel stats, geospatial metadata retained per run

Every adapted model benchmarked against the base model before promotion

Early Evidence

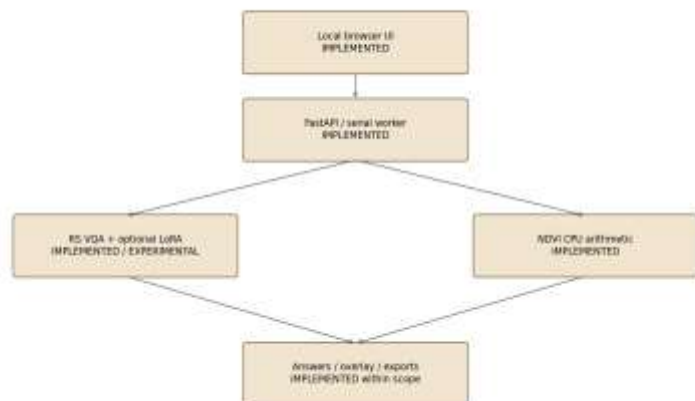
Single-image VQA and calibrated NDVI validated in local experiments · ~4.3 GiB peak GPU usage · development accuracy improved 38.3% → 78.3% across 60 questions/20 images

(development-only result;
formal benchmark evaluation still pending)

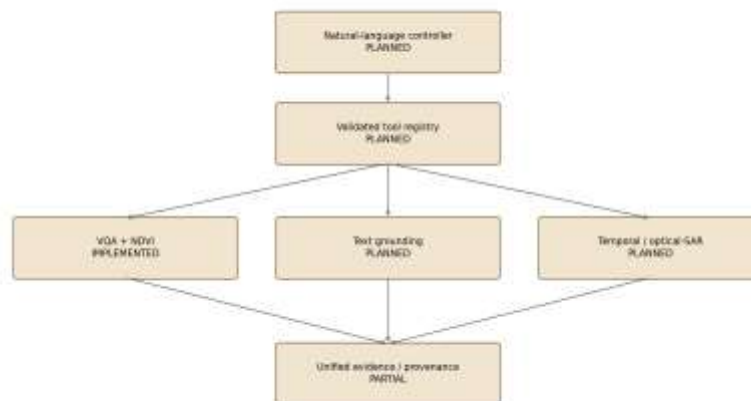


EXPERIMENTAL VALIDATION

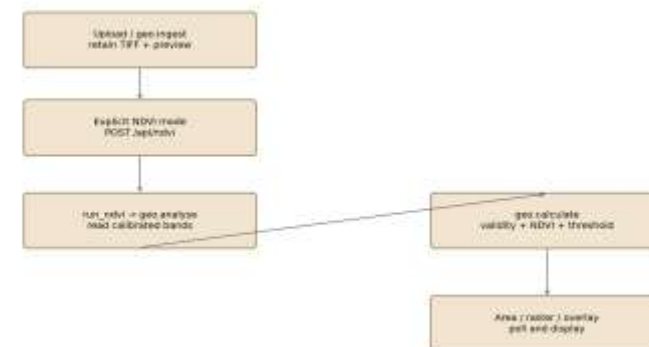
CURRENT — explicit modes, not automatic routing



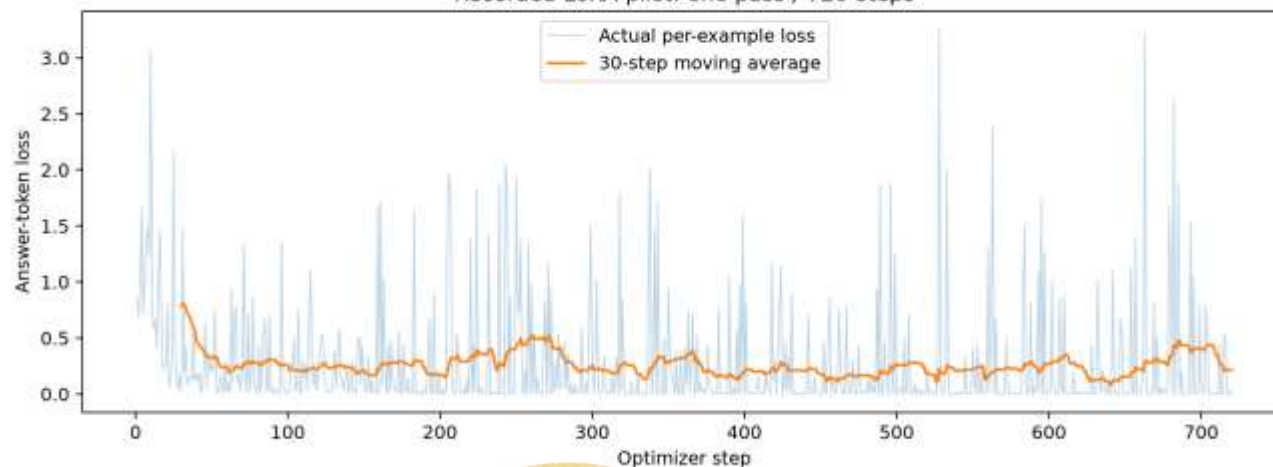
TARGET — planned components are explicitly labelled



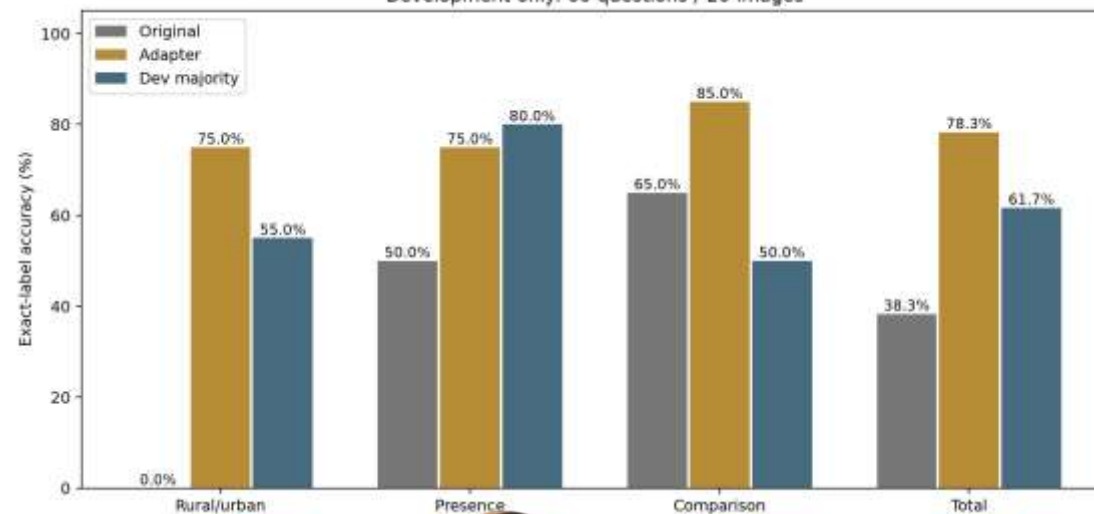
ACTUAL NDVI FLOW — manual selection



Recorded LoRA pilot: one pass / 720 steps



Development only: 60 questions / 20 images



(development-only result;
formal benchmark evaluation still pending)

IMPACT AND BENEFITS

Impact on the Target Audience

Researchers & Students: Query satellite imagery in plain language instead of needing GIS training to get started.

GIS & Remote-Sensing Professionals: One interface for semantic questions and quantitative analysis, cutting down tool-switching.

Government & Planning Bodies: Faster preliminary reads on urban, agricultural, and environmental data.

Disaster-Response Teams: Once temporal and SAR capabilities are built, they'll support rapid before/after comparisons.

Non-GIS Experts: Ask real questions of satellite data without building a full GIS pipeline first.

Benefits

👥 Social

Opens up Earth observation beyond trained specialists

Turns complex imagery into answers, numbers, and visual evidence people can actually use

Natural language lowers the barrier for students, researchers, and public-sector users

🔧 Technological

Pairs a VLM's semantic reasoning with deterministic math, so numbers are computed, not guessed

Every result can carry overlays, pixel stats, thresholds, and metadata as evidence

Built to expand - grounding, temporal analysis, and optical-SAR fusion can be added as new specialists

LoRA/PEFT adapts the model to remote sensing without retraining it from scratch

💰 Economic / Operational

Runs on an 8GB consumer GPU - no expensive hardware required to experiment or deploy

One interface replaces switching between separate tools for interpretation and analysis

Open-source stack (Python, FastAPI, PyTorch) - no proprietary dependencies

Can move from local deployment to cloud GPU infrastructure if scale demands it

🌍 Environmental

NDVI gives a measurable read on vegetation patterns today

Supports interpreting land cover and visible environmental change

Temporal analysis, once built, can track how regions shift over time

Planned optical-SAR support extends observation into conditions optical imagery can't handle alone

"SatQuery turns satellite imagery from something users must learn to operate into something they can directly interrogate, while keeping measurements computational, traceable and inspectable."



RESEARCH & REFERENCES

Research Foundation

Remote-Sensing Vision-Language Models:

Prior work shows that domain-specific vision-language models can interpret overhead imagery, answer remote-sensing questions, and support spatial reasoning. **GeoChat**, **RemoteCLIP** and **EarthVQA** establish the research basis for conversational and queryable Earth observation.

Domain-Adaptive Multimodal Models:

SatQuery uses **AdaptLLM Remote-Sensing Qwen2-VL-2B-Instruct** as its starting VLM. The checkpoint is derived from Qwen2-VL and post-trained specifically for remote sensing using domain visual instructions.

Parameter-Efficient Adaptation:

Our adaptation follows the **LoRA** principle: preserve the pretrained model while training small low-rank parameters, reducing the resources required for domain/task adaptation.

Grounding & Future Specialists:

Research such as **GeoChat** demonstrates that remote-sensing VLMs can extend beyond image-level answers toward region-specific dialogue and spatial grounding-supporting SatQuery's planned grounding specialist.

Key References

[1] Cheng et al. - Domain-Specific Post-Training for Multimodal LLMs

Basis for the remote-sensing-adapted Qwen2-VL checkpoint used by SatQuery.

[2] Hu et al. - LoRA: Low-Rank Adaptation of Large Language Models

Foundation for SatQuery's parameter-efficient adaptation approach.

[3] Kuckreja et al. - GeoChat, CVPR 2024

Remote-sensing VLM supporting VQA, region dialogue and visual grounding.

[4] Liu et al. - RemoteCLIP

Vision-language foundation-model research specifically targeting remote-sensing imagery.

[5] Wang et al. - EarthVQA, AAAI 2024

Remote-sensing VQA benchmark focused on relational reasoning, counting and Earth-scene analysis.

[6] AdaptLLM - Remote-Sensing Visual Instructions

36K+ image/text examples derived from multiple established remote-sensing caption datasets and used for domain-specific post-training.

AdaptLLM Remote-Sensing Qwen2-VL-2B-Instruct — model actually used by SatQuery

<https://huggingface.co/AdaptLLM/remote-sensing-Qwen2-VL-2B-Instruct>

AdaptLLM Remote-Sensing Visual Instructions — upstream domain-training dataset

<https://huggingface.co/datasets/AdaptLLM/remote-sensing-visual-instructions>

On Domain-Adaptive Post-Training for Multimodal Large Language Models — research behind AdaptLLM checkpoint

<https://arxiv.org/abs/2411.19930>

LoRA: Low-Rank Adaptation of Large Language Models — Hu et al.

<https://arxiv.org/abs/2106.09685>

GeoChat: Grounded Large Vision-Language Model for Remote Sensing — CVPR 2024

https://openaccess.thecvf.com/content/CVPR2024/html/Kuckreja_GeoChat_Grounded_Large_Vision-Language_Model_for_Remote_Sensing_CVPR_2024_paper.html

GeoChat official repository

<https://github.com/mbzuai-oryx/GeoChat>

RemoteCLIP: A Vision Language Foundation Model for Remote Sensing

<https://arxiv.org/abs/2306.11029>

EarthVQA: Towards Queryable Earth via Remote-Sensing VQA — AAAI 2024

<https://ojs.aaai.org/index.php/AAAI/article/view/28357>